



SAGI RAMA KRISHNAM RAJU ENGINEERING COLLEGE (AUTONOMOUS)

(Approved by AICTE, New Delhi, Affiliated to JNTUK, Kakinada)

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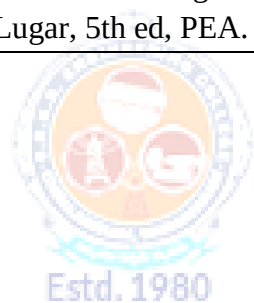
SRKR MARG, CHINA AMIRAM, BHIMAVARAM – 534204 W.G.Dt., A.P., INDIA

LIST OF OPEN ELECTIVES OFFERED BY VARIOUS DEPARTMENTS TO OTHER DEPARTMENTS IN IV YEAR I SEMESTER **OE-III**

Offered by	Course Code	Course Name	Offered to
ARTIFICIAL INTELLIGENCE & DATA SCIENCE	B23ADOE05	Principles of Artificial Intelligence	CE, ECE, EEE & ME
ARTIFICIAL INTELLIGENCE & MACHINE LEARNING	B23AMOE06	Deep Learning	CE, ECE, EEE & ME
	B23AMOE07	Object Oriented Programming Through Java	
CIVIL ENGINEERING	B23CEOE05	Sustainability in Engineering practices	AIDS, AIML, CIC, CSBS, CSE, CSG, CSIT, ECE, EEE, IT & ME
	B23CEOE06	Solid waste Management	
COMPUTER SCIENCE & BUSINESS SYSTEMS	B23CBOE04	Digital Marketing	CE, ECE, EEE & ME
COMPUTER SCIENCE & ENGINEERING	B23CSOE04	Object Oriented Programming through Java	CE, ECE, EEE & ME
CSE (Internet of Things and Cyber Security including Block Chain Technology)	B23CIOE05	Introduction to Cryptography & Network Security	CE, ECE, EEE & ME
	B23CIOE06	Computer Networks	
ELECTRONICS & COMMUNICATION ENGINEERING	B23ECOE03	Optical Communication	AIDS, AIML, CE, CIC, CSBS, CSE, CSG, CSIT, EEE, IT & ME
	B23ECOE04	Fundamentals of VLSI Design	
	B23ECOE05	Electronic Measurements and Instrumentation	AIDS, AIML, CE, CIC, CSBS, CSE, CSG, CSIT, IT & ME
ELECTRICAL & ELECTRONICS ENGINEERING	B23EEOE05	Autonomous Vehicle Systems	AIDS, AIML, CE, CIC, CSBS, CSE, CSG, CSIT, ECE, IT & ME
	B23EEOE06	Electrical Circuit Analysis	AIDS, AIML, CE, CIC, CSBS, CSE, CSG, CSIT, IT & ME
INFORMATION TECHNOLOGY	B23ITOE03	Principles Of Database Management Systems	AIML, CIC OE-II
MECHANICAL ENGINEERING	B23MEOE07	Total Quality Management	AIDS, AIML, CE, CIC, CSBS, CSE, CSG, CSIT, ECE, EEE & IT
	B23MEOE08	Smart Manufacturing	
MATHEMATICS AND HUMANITIES	B23BSOE03	Mathematical Modelling For Data Science	AIDS, AIML, CE, CIC, CSBS, CSE, CSG, CSIT, ECE, EEE, IT & ME

Course Code	Category	L	T	P	C	C.I.E.	S.E.E.	Exam
B23ADOE05	OE	3	--	--	3	30	70	3 Hrs.
PRINCIPLES OF ARTIFICIAL INTELLIGENCE								
Offered by AIDS								
(Offered to CE, ECE, EEE &ME)								
Course Objectives: This course is designed to:								
1	To have a basic proficiency in a traditional AI language including an ability to write simple to intermediate programs and an ability to understand code written in that language							
2	To understand the basic issues of knowledge representation and blind and heuristic search, as well as an understanding of other topics such as minimax, resolution that play an important role in AI programs							
3	To have a basic understanding of some of the more advanced topics of AI							
Course Outcomes: Upon the completion of the course students will be able to:								
S.No	Outcome							Knowledge Level
1	Describe the basic foundations & applications of AI.							K2
2	Apply problem solving strategies to generate the best AI solutions using state space search.							K3
3	Use AI languages to represent knowledge base for problem solving							K3
4	Use AI tools to represent knowledge base in real world problems.							K3
5	Apply uncertainty and fuzzy logic techniques to solve AI problems							K3
SYLLABUS								
UNIT-I (10Hrs)	Introduction, history, intelligent systems, foundations of AI, applications, tic-tac-toe game playing, development of AI languages, current trends.							
UNIT-II (8 Hrs)	Problem solving: state-space search and control strategies: Introduction, general problem solving, characteristics of problem, exhaustive searches, heuristic search techniques, iterative deepening A*, constraint satisfaction Problem reduction and game playing: Introduction, problem reduction, game playing, alpha beta pruning, two-player perfect information games.							
UNIT-III (10 Hrs)	Logic concepts: Introduction, propositional calculus, propositional logic, natural deduction system, axiomatic system, semantic tableau system in propositional logic, resolution refutation in propositional logic, predicate logic.							
UNIT-IV (10 Hrs)	Knowledge representation: Introduction, approaches to knowledge representation, knowledge representation using semantic networks, extended semantic networks for KR,							

	knowledge representation using frames Advanced knowledge representation techniques: Introduction, conceptual dependency theory, script structure
UNIT-V (8 Hrs)	Expert system and applications: Introduction phases in building expert systems, expert system versus traditional systems Uncertainty measure: probability theory: Introduction, probability theory, Bayesian belief networks Fuzzy sets and fuzzy logic: Introduction, fuzzy sets, fuzzy set operations, types of membership functions
Textbooks:	
1.	Artificial intelligence, A modern Approach, 2nded, Stuart Russel, Peter Norvig, PEA.
2.	Artificial Intelligence- Deepak Khemani, TMH, 2013.
Reference Books:	
1.	Artificial Intelligence- Saroj Kaushik, CENGAGE Learning.
2.	Introduction to Artificial Intelligence, Patterson, PHI.
3.	Artificial intelligence, structures and Strategies for Complex problem solving, George F Lugar, 5th ed, PEA.



Course Code	Category	L	T	P	C	C.I.E.	S.E.E.	Exam
B23AMOE06	OE	3	--	--	3	30	70	3 Hrs.
DEEP LEARNING								
Offered by AIML								
(Offered to CE, ECE, EEE &ME)								
Pre-requisites: Nil								
Course Objectives: This course aims to equip students with the following:								
1	Understand and recollect basic concepts of machine learning							
2	Understand concepts of deep feed forward network mechanisms							
3	Understand and analyze the concepts of CNN, RNN models							
4	Study the concepts of auto encoders and optimization techniques							
5	Study and analyze the different DNN architectures							
Course Out Comes: At the end of the course students will be able to								
S.No	Outcome							Knowledge Level
1	Demonstrate the basic concept of Machine learning							K3
2	Apply the concepts of deep feed forward networks.							K3
3	Apply the concepts of CNN & RNN models							K3
4	Apply autoencoders for feature extraction and optimization algorithms for training deep learning models.							K3
5	Apply deep learning architectures and models to tasks like image classification, sentiment analysis and object detection							K3
SYLLABUS								
UNIT-I (12 Hrs)	Fundamentals Concepts of Machine Learning Historical Trends in Deep Learning-Machine Learning Basics: Learning Algorithms- Supervised and Unsupervised Training, Linear Algebra for machine Learning, Testing, Cross-Validation, Dimensionality reduction, Over/Under-fitting, Hyper parameters and validation sets, Bias, Variance, Regularization							
UNIT-II (10 Hrs)	Deep Feed Forward Networks Deep feed forward networks-Introduction, Various Activation Functions, error functions- Regularization for Deep learning-L1,L2 regularization, Data augmentation, Early Stopping, Drop out.							
UNIT-III (10 Hrs)	Convolutional Neural Networks And Sequence Modeling Convolutional Networks: Convolutional operation- Pooling- Normalization, Sequence Modeling: Recurrent Neural Networks, The Long Short-Term Memory.							

UNIT-IV (08 Hrs)	Auto Encoders and Optimization Algorithms Auto encoders - Auto encoders: under complete, denoising, Optimization for Deep Learning: gradient descent, stochastic gradient descent, mini batch gradient descent, Adagrad, RMSProp, Adam
UNIT-V (10 Hrs)	More Deep Learning Architectures & Applications Alexnet, VGGNet, ResNet, Transfer learning, Applications: Sentiment Analysis using LSTM, Object Detection
TEXTBOOK:	
1.	Ian Goodfellow, Yoshua Bengio, Aaron Courville, “Deep Learning”, MIT Press, 2016 (available at http://www.deeplearningbook.org)
2.	Charu C Agarwal, “Neural Networks and Deep Learning”, IBM T. J. Watson Research Center, International Business Machines, Springer, 2018
REFERENCE BOOKS:	
1.	Michael Nielsen, “Neural Networks and Deep Learning”, Online book, 2016 (http://neuralnetworksanddeeplearning.com/)
2.	Jason Brownlee , “Deep Learning with Python” , ebook, 2016
3.	N. D. Lewis, “Deep Learning Step by Step with Python: A Very Gentle Introduction to Deep Neural Networks for Practical Data Science, 2016.
4.	Chris Albon, “Machine Learning with Python Cookbook-practical solutions from preprocessing to Deep learning”, O'REILLY Publisher,2018
Useful Reference Links:	
1.	https://www.geeksforgeeks.org/activation-functions-neural-networks/
2.	https://towardsdatascience.com/sentiment-analysis-using-lstm-step-by-step-50d074f09948
3.	https://medium.com/@lamiae.hana/a-step-by-step-guide-on-sentiment-analysis-with-rnn-and-lstm-3a293817e314
4.	https://towardsdatascience.com/common-loss-functions-in-machine-learning-46af0ffc4d23
5.	https://d2l.ai/chapter_natural-language-processing-applications/sentiment-analysis-rnn.html
Web Link:	
1.	Swayam NPTEL: Deep Learning: https://onlinecourses.nptel.ac.in/noc22_cs22/preview

Course Code	Category	L	T	P	C	C.I.E.	S.E.E.	Exam
B23AMOE07	OE	3	--	--	3	30	70	3 Hrs.
OBJECT ORIENTED PROGRAMMING THROUGH JAVA								
Offered by AIML								
(Offered to CE, ECE, EEE & ME)								
Course Objectives: This course aims to equip students with the following:								
1.	To identify Java language components and how they work together in applications.							
2.	To learn the fundamentals of object-oriented programming in Java, including defining classes, invoking methods, using class libraries.							
3.	To learn how to extend Java classes with inheritance and dynamic binding and how to use exception handling, file I/O in Java applications.							
4.	To understand how to design applications with threads in Java.							
Course Outcomes: At the end of the course students will be able to								
S. No	Outcome							Knowledge Level
1.	Apply basic Object-Oriented Programming Concepts in Java.							K3
2.	Use Arrays and Strings to solve real world problems.							K3
3.	Apply the concepts of Inheritance and Interfaces for developing reusable programs.							K3
4.	Apply the concepts of packages and exception handling to develop modular and error-free Java applications.							K3
5.	Apply Multithreading and file I/O operations to develop efficient and concurrent Java applications.							K3
SYLLABUS								
UNIT-I (10 Hrs)	Introduction to OOP: Basic concepts of OOP and Advantages of OOP. Introduction to JAVA: Structure of JAVA, Features of JAVA, Data Types, JAVA Tokens, Control Statements. Classes & Objects: Introduction, Class Declaration, Class Members, Declaration of Class Objects, Static Variables and Methods, User input to programs, Command Line Arguments, Method overloading. Constructors: Default Constructor, Parameterized Constructor, Copy Constructor and Constructor Overloading, This Keyword.							
UNIT-II (08 Hrs)	Arrays: Introduction, Declaration and Initialization of Arrays, Accessing Elements of Arrays, Two-dimensional Arrays and Variable Size Arrays. String Handling in Java: Introduction, Methods in String class, String Constant Pool and String Buffer class, Wrapper classes, Type Conversion.							

UNIT-III (10 Hrs)	Inheritance: Introduction, Single inheritance, Multi-level inheritance, Hierarchical Inheritance, Method Overriding, Super Keyword, Final Keyword and Abstract Classes. Interfaces: Introduction, Declaration of Interface, Implementing Multiple Inheritance, Extending interfaces.
UNIT-IV (10 Hrs)	Packages: Introduction, Defining Package, Java util Classes and Interfaces, Java lang Classes, Importing Packages, Sub Packages, Access Modifiers. Exception Handling: Introduction, Hierarchy of Standard Exception Classes, Keywords try, catch, throw, throws and finally Blocks, Multiple Catch Statements, Custom Exceptions, Nested try and catch Blocks.
UNIT-V (10 Hrs)	Multithreaded Programming: Introduction, Thread Life Cycle, Extending Thread class, Implementing Runnable interface, Thread Priorities, Thread Synchronization. File IO: Introduction, Hierarchy of Stream classes, Byte Streams, Character streams.
Textbooks:	
1.	The complete Reference Java, 12th edition (2022), Herbert Schildt, Publisher: McGraw Hill.
2.	JdbcApi Tutorial and Reference 3E (2003), by Maydene, Jon Ellis (Author), Jonathan Bruce (Author), Publisher: Addison-Wesley Professional.
3.	Joy with JAVA, Fundamentals of Object-Oriented Programming, Debasis Samanta, Monalisa Sarma, Cambridge, 2023.
Reference Books:	
1.	Introduction to java programming, 9th edition (2014) by Y Daniel Liang, Publisher: Pearson.
2.	Murach's Java Programming, 5th edition (2017) Joel Murach, Publisher: Mike Murach.
3.	JAVA one step ahead, 1st edition (2017) Anitha Seth, B.L.Juneja, Oxford.
4.	Java: A Beginner's Guide, Eighth Edition 8th Edition (2018) by Herbert Schildt, Publisher: McGraw-Hill Education.
5.	Head First Java 3e (2021) (A Brain Friendly Guide) by Kathy Sierra & Bert bates, Publisher: O'Reilly.
6.	Programming with Java: A Primer 6E (2019) By Balagurusamy, Publisher: TMH.
e-Resources	
1.	https://nptel.ac.in/courses/106/105/106105191/
2.	https://www.coursera.org/learn/java-introduction
3.	https://docs.oracle.com/javase/tutorial/

Course Code	Category	L	T	P	C	C.I.E.	S.E.E.	Exam
B23CEOE05	OE	3	--	--	3	30	70	3 Hrs.
SUSTAINABILITY IN ENGINEERING PRACTICES								
(Offered by CE)								
(Offered to AIDS, AIML, CIC, CSBS, CSE, CSG, CSIT, ECE, EEE, IT & ME)								
Course Objectives:								
1.	To introduce the fundamental concepts of sustainability, including sustainable development, environmental ethics, and global environmental issues affecting natural resources.							
2.	To provide knowledge on environmental management tools and practices, such as Environmental Management Systems (EMS), Life Cycle Assessment (LCA), and Environmental Impact Assessment (EIA).							
3.	To familiarize students with sustainable design and development practices, including green buildings, sustainable habitats, urbanization, and industrial sustainability approaches.							
4.	To develop an understanding of renewable energy resources and green technologies, and their role in achieving sustainable engineering solutions.							
Course Outcomes								
S.No	Outcome							Knowledge Level
1.	Explain sustainable development and different environmental agreements and protocols							K2
2.	Discuss real time activities causing environmental issues and different methods to use renewable energy resources							K2
3.	Explain local and global environmental issues							K2
4.	Differentiate between carbon emissions for regular and sustainable cities and explain different practices to move industries towards sustainability							K3
5.	Discuss different renewable energy resources and explain methods to implement green technology							K2
SYLLABUS								
UNIT-I (10 Hrs)	Introduction to Sustainable Engineering- Sustainable development, concepts of sustainable development: three pillar model, egg of sustainability model, Atkisson's pyramid model, prism model, principles of sustainable development, sustainable engineering, threats for sustainability. Environmental Ethics and Legislations – Environmental ethics and education, multilateral environmental agreements and protocols, enforcement of environmental laws in India – The Water Act, The Air Act, The Environment Act.							
UNIT-II (8Hrs)	Local Environmental Issues- Solid waste, impact of solid waste on natural resources, zero waste concept and three R concept, waste to energy technology: thermo-chemical							

	conversion, biochemical conversion. Global Environmental Issues- Resource degradation: deterioration of water resources, land degradation, air pollution, climate change and global warming, ozone layer depletion, carbon footprint, carbon trading.
UNIT-III (8Hrs)	Tools for Sustainability - Environmental management System (EMS), concept of ISO14000, life cycle assessment (LCA): basic components, advantages, disadvantages, case study. Environmental impact assessment (EIA), environmental auditing, bio mimicking, case studies.
UNIT-IV (10 Hrs)	Sustainable Habitat - Concept of green building, green building materials, green building certification and rating: green rating for integrated habitat assessment(GRIHA), leader ship in energy and environmental design (LEED) rating, energy efficient buildings, sustainable cities, sustainable transport, sustainable pavements, case studies in sustainability engineering: Green building, sustainable city, sustainable transport system. Sustainable Industrialization and Urbanization – Sustainable urbanization, industrialization, material selection, pollution prevention, industrial ecology, industrial symbiosis, poverty reduction.
UNIT-V (8Hrs)	Renewable energy resources- Conventional and non- conventional forms of energy, solar energy, fuel cells, wind energy, small hydroplants, biogas systems, biofuels, energy from ocean, geothermal energy, conservation of energy. Green technology and Green Business: Sustainable business, green technology, green energy, green construction, green transportation, green chemistry, green computing
Textbooks:	
1.	R.L. Ragand Lekshmi Dinachandran Remesh. <i>Introduction to Sustainable Engineering</i> . 2nd Edition, PHI Learning Pvt. Ltd., 2016.
2.	D.T.Allen and D.R.Shonnard. <i>Sustainability Engineering: Concepts, Design and Case Studies</i> , 1st Edition, Prentice Hall, 2011.
Reference Books:	
1.	A.S.Bradley, A.O.Adebayo, P.Maria. <i>Engineering applications in sustainable design and development</i> , 1st Edition, Cengage learning, 2016.
e-Resources	
1.	https://onlinecourses.nptel.ac.in/noc22_ce07/preview

Course Code	Category	L	T	P	C	C.I.E.	S.E.E.	Exam
B23CEO06	OE	3	--	--	3	30	70	3 Hrs.
SOLID WASTE MANAGEMENT								
(Offered by CE)								
(Offered to AIDS, AIML, CIC, CSBS, CSE, CSG, CSIT, ECE, EEE, IT & ME)								
Course Objectives:								
1.	To familiarize with sources, types and characteristics of solid waste.							
2.	To impart knowledge on collection, transportation, processing and resource recovery.							
3.	To understand design and operation of waste disposal systems and landfills.							
4.	To introduce hazardous waste management practices and regulations.							
Course Outcomes								
S.No	Outcome							Knowledge Level
1.	Categorize and explain the types of solid wastes along with their sources and characteristics.							K2
2.	Describe collection, transportation, transfer and processing methods for solid waste management.							K2
3.	Elaborate processing and disposal methods of solid waste including recovery and treatment techniques.							K3
4.	Illustrate landfill design, operation, leachate management and closure methods.							K3
5.	Outline hazardous waste management practices, treatment methods and regulations.							K2
SYLLABUS								
UNIT-I (10Hrs)	Introduction: Definition of solid waste – waste generation-sources and types of solid waste – sampling and characterization – Determination of composition of Municipal Solid Waste – Onsite storage and handling of solid waste – Future changes in waste composition, major legislation, monitoring responsibilities.							
UNIT-II (10 Hrs)	Collection and Transport of Solid Waste: Type and methods of waste collection systems, analysis of collection systems-Optimization of collection routes– alternative routes for collection system. Transfer and Transport: Need for transfer operation, transport means and methods, transfer station types. Separation and Processing and Transformation of Solid Waste- Waste as a Resource- Disposable Materials, Recycling, Collection, Processing, and Potential for Reuse.							
UNIT-III (10 Hrs)	Processing and disposal: unit operations used for separation and processing, Materials Recovery facilities, Source reduction and waste minimization, Metal Separation &							

	Recovery Waste transformation through combustion and composting, anaerobic methods for materials recovery and treatment – Energy recovery – biogas generation and cleaning– Incinerators.
UNIT-IV (10 Hrs)	Landfills: Classification, Site selection, design and operation, methods of filling, drainage and leachate collection systems –designated waste landfill remediation, reclamation, environmental closure.
UNIT-V (10 Hrs)	Hazardous Waste Management: Definition and identification of hazardous wastes-sources and characteristics – hazardous wastes in Municipal Waste – Hazardous wastes regulations – minimization of Hazardous Waste-compatibility, handling and storage of hazardous waste collection and transport, e-waste - sources, collection. Design and operation of facilities for physical, chemical and thermal treatment of hazardous waste – Solidification, chemical fixation and encapsulation, incineration – reuse after treatment.
Textbooks:	
1.	Solid and Hazardous Waste Management, 2nd Edition, M.N.Rao, Razia Sultana, BS Publications / BSP Books; 2nd edition (1 January 2020)
2.	Solid Waste Management, K Sasi Kumar, Prentice Hall India Learning Private Limited (1 January 2009)
Reference Books:	
1.	George Tchobanoglous “Integrated Solid Waste Management”, McGraw Hill Publication, 1993.
2.	Charles A. Wentz; “Hazardous Waste Management”, McGraw Hill Publication, 19.
3.	Solid And Hazardous Waste Management, P.M.Cherry, CBS Publishers (1 January 2017)
4.	Integrated Solid Waste Management by George Tchobanoglous, McGraw Hill Education (28 April 2014)
e-Resources	
1.	NPTEL – Solid Waste Management Course
2.	SWAYAM – Environmental Engineering & Solid Waste Management

Course Code	Category	L	T	P	C	C.I.E.	S.E.E.	Exam
B23CSOE04	OE	3	--	--	3	30	70	3 Hrs
DIGITAL MARKETING								
(Offered by CSBS)								
(Offered to CE, ECE, EEE & ME)								
Course Objectives: This course is designed to:								
1.	Understand the fundamental concepts and evolution of digital marketing in modern business environments.							
2.	Analyze digital advertising models, formats, and metrics for effective campaign performance evaluation.							
3.	Apply knowledge of search engine optimization (SEO) and search engine advertising (SEA) techniques to improve online visibility.							
4.	Design and implement social media marketing strategies across various platforms.							
5.	Understand Indian Aspect of Digital Marketing							
Course Outcomes: Upon the completion of the course students will be able to:								
S.No	Outcome							Knowledge Level
1.	Relate the Basics of Digital Marketing in the Modern Era							K3
2.	Apply the Display Advertising Concepts							K3
3.	Prepare for the Search Engine Optimization/Advertising							K3
4.	Prepare for the Tactics of Social Media Marketing							K3
5.	Predict the Scope of Digital Marketing in Indian Context							K2
SYLLABUS								
UNIT-I (10Hrs)	Basics of Digital Marketing Definition of Digital Marketing-Evolution of Digital Marketing: from traditional to modern marketing, the rise of the internet: The Dotcom Era, Post Dotcom: Creation of Internet Business Models-emergence of digital marketing as a new tool-Application of Digital Marketing							
UNIT-II (8 Hrs)	Display Advertising Concept of Display Advertising: Display Advertising Media, Digital Metrics-Types of Display Ads: Format, Display Ad Size- Buying Models: Cost per Click (CPC), Cost per Milli (CPM), Cost per Lead (CPL), Cost per Acquisition (CPA), Fixed cost/ Sponsorship.							
UNIT-III (10 Hrs)	Search Engine Advertising and Search Engine Optimization Introduction about how search engine works, why pay for Search Advertising: Capture							

	Intent, Ease of Action, Controlled Costs, Analytics, Competition-Understanding Ad Placements: Top, Side, Bottom-Concept of Search Engine Optimization, Important elements to be considered in Search Engine Optimization.
UNIT-IV (10 Hrs)	Social Media Marketing Introduction to Social Media Marketing- Building a Successful Strategy: Listen, Goal Setting, Strategy, Implementation, Measure, Improve-Facebook Marketing-LinkedIn Marketing-Twitter Marketing-Instagram and Snapchat.
UNIT-V (8 Hrs)	Digital Marketing - The Indian View India Digital Spend Overview, Emerging trends of Digital Marketing: Big Data and IoT- Data technologies Impacting Marketing, B2B and SMB – Segments based Digital Marketing, SoLoMo – the Next level of Hyperlocal Marketing
Textbooks:	
1.	Fundamentals of Digital Marketing by Puneet Singh Bhatia, Pearson Publishers, 2 nd Edition 2019
2.	Digital Marketing by Seema Gupta, Mc GrawHill Publishers, 2 nd Edition 2020
Reference Books:	
1.	Digital Marketing by Dave Chaffey and Fiona Ellis- Chadwick by Pearson Publishers, 2019
2.	Applications of Digital Marketing for success in Business by Abhishek Das, BPB Publications, 2018.
3.	Fundamentals of Digital Marketing, Sun India, 2020.
4.	Digital Marketing Essentials you always wanted to know, Vibrant Publishers, 2020

Course Code	Category	L	T	P	C	C.I.E.	S.E.E.	Exam
B23CSOE04	OE	3	--	--	3	30	70	3 Hrs.
OBJECT ORIENTED PROGRAMMING THROUGH JAVA								
(Offered by CSE)								
(Offered to CE, ECE, EEE & ME)								
Course Objectives: The main objectives of the course is to								
1.	To identify Java language components and how they work together in applications.							
2.	To learn the fundamentals of object-oriented programming in Java, including defining classes, Invoking methods, using class libraries.							
3.	To learn how to extend Java classes with inheritance and dynamic binding and how to use Exception handling, file I/O in Java applications.							
4.	To understand how to design applications with threads in Java.							
Course Outcomes: At the end of the course Students will be able to								
S. No	Outcome							Knowledge Level
1.	Apply basic Object-Oriented Programming Concepts in Java.							K3
2.	Use Arrays and Strings to solve real world problems.							K3
3.	Apply the concepts of Inheritance and Interfaces for developing reusable programs.							K3
4.	Apply the concepts of packages and exception handling to develop modular and error-free Java applications.							K3
5.	Apply Multithreading and file I/O operations to develop efficient and concurrent Java applications.							K3
SYLLABUS								
UNIT-I (10 Hrs)	Introduction to OOP: Basic concepts of OOP and Advantages of OOP. Introduction to JAVA: Structure of JAVA, Features of JAVA, Data Types, JAVA Tokens, Control Statements. Classes & Objects: Introduction, Class Declaration, Class Members, Declaration of Class Objects, Static Variables and Methods, User input to programs, Command Line Arguments, Method overloading. Constructors: Default Constructor, Parameterized Constructor, Copy Constructor and Constructor Overloading, This Keyword.							
UNIT-II (08 Hrs)	Arrays: Introduction, Declaration and Initialization of Arrays, Accessing Elements of Arrays, Two-dimensional Arrays and Variable Size Arrays. String Handling in Java: Introduction, Methods in String class, String Constant Pool							

	and String Buffer class, Wrapper classes, Type Conversion.
UNIT-III (10 Hrs)	Inheritance: Introduction, Single inheritance, Multi-level inheritance, Hierarchical Inheritance, Method Overriding, Super Keyword, Final Keyword and Abstract Classes. Interfaces: Introduction, Declaration of Interface, Implementing Multiple Inheritance, Extending interfaces.
UNIT-IV (10 Hrs)	Packages: Introduction, Defining Package, Java util Classes and Interfaces, Java lang Classes, Importing Packages, Sub Packages, Access Modifiers. Exception Handling: Introduction, Hierarchy of Standard Exception Classes, Keywords try, catch, throw, throws and finally Blocks, Multiple Catch Statements, Custom Exceptions, Nested try and catch Blocks.
UNIT-V (10 Hrs)	Multithreaded Programming: Introduction, Thread Life Cycle, Extending Thread class, Implementing Runnable interface, Thread Priorities, Thread Synchronization. File IO: Introduction, Hierarchy of Stream classes, Byte Streams, Character streams.
Textbooks:	
1.	The complete Reference Java, 12th edition (2022), Herbert Schildt, Publisher: McGraw Hill.
2.	JdbcApi Tutorial and Reference 3E (2003), by Maydene, Jon Ellis (Author), Jonathan Bruce (Author), Publisher: Addison-Wesley Professional.
3.	Joy with JAVA, Fundamentals of Object-Oriented Programming, Debasis Samanta, Monalisa Sarma, Cambridge, 2023.
Reference Books:	
1.	Introduction to java programming, 9th edition (2014) by Y Daniel Liang, Publisher: Pearson.
2.	Murach's Java Programming, 5th edition (2017) Joel Murach, Publisher: Mike Murach.
3.	JAVA one step ahead, 1st edition (2017) Anitha Seth, B.L.Juneja, Oxford.
e-Resources:	
1.	https://onlinecourses.nptel.ac.in/noc25_cs57/preview
2.	https://www.coursera.org/learn/java-introduction
3.	https://docs.oracle.com/javase/tutorial/

Course Code	Category	L	T	P	C	I.M	E.M	Exam
B23CIOE05	OE	3	--	--	3	30	70	3Hrs.
INTRODUCTION TO CRYPTOGRAPHY AND NETWORK SECURITY								
(Offered by CIC)								
(Offered to CE, ECE, EEE & ME)								
Pre-requisites: Computer Networks, Data Communications								
Course Objectives:								
1.	Overview of computer security and classical encryption techniques.							
2.	Working principles and utilities of various cryptographic algorithms including symmetric key cryptography and public key cryptography algorithms.							
3.	Design issues and working principles of hashing, message digest algorithms and various authentication protocols.							
4.	Various secure communication protocols standards and Concepts of firewalls.							
Course Outcomes: At The End Of the course students will be able								
								Knowledge Level
1.	Illustrate various types of Security attacks and Block Cipher design principles.							K2
2.	Apply Symmetric Key Cryptography techniques & Public Key Cryptography Algorithms to achieve confidentiality service.							K3
3.	Use Hashing algorithms, Digital algorithms to achieve Integrity service.							K3
4.	Describe User Authentication and Email, IP security protocols.							K2
5.	Explain Transport level security protocols and the Importance of system security through firewalls.							K2
SYLLABUS								
UNIT-I (8 Hrs)	Classical Encryption Techniques: Security Attacks, Services & Mechanisms, Symmetric Cipher Model. Phishing Attack, Web Based Attacks, SQL Injection Attacks. Block Ciphers: Traditional Block Cipher Structure, Block Cipher Design Principles.							
UNIT-II (10 Hrs)	Symmetric Key Cryptography: Data Encryption Standard (DES), Advanced Encryption Standard (AES), IDEA, Block Cipher Modes of Operations. Public Key Cryptography: Principles, Public Key Cryptography Algorithms, RSA Algorithm, Diffie Hellman Key Exchange.							

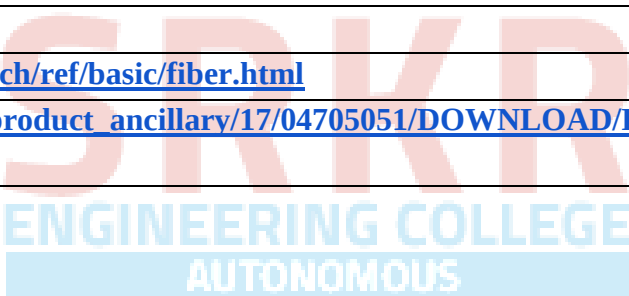
UNIT-III (12 Hrs)	Cryptographic Hash Functions: Application of Cryptographic Hash Functions, Requirements & Security, SHA-512, MD5, Message Authentication Functions, Requirements & Security, HMAC & CMAC. Digital Signatures: Digital Signature Algorithm, DSS with RSA.
UNIT-IV (10 Hrs)	User Authentication: Remote User Authentication Principles, Kerberos. Electronic Mail Security: Pretty Good Privacy (PGP) And S/MIME. IP Security: IP Security Overview, IP Security Architecture, Authentication Header, Encapsulating Security Payload.
UNIT-V (10Hrs)	Transport Level Security: Web Security Requirements, Secure Socket Layer (SSL) and Transport Layer Security (TLS) Firewalls: Characteristics, Types of Firewalls, Placement of Firewalls, Firewall Configuration, Trusted Systems.
Text Books:	
1.	Cryptography and Network Security – Principles and Practice, William Stallings, 7/e. Pearson Education, 2017
2.	Cryptography and Network Security- Behrouz A Forouzan, Debdeep Mukhopadhyaya, McGrawHill, 3rd Edition, 2015
Reference Books:	
1.	Neal Koblitz, A Course in Number Theory and Cryptography: Springer- Verlag, New York Inc. May 2001.
2.	Atul Kahate, Cryptography and Network Security, 4/e, McGraw Hill, 2019

Course Code	Category	L	T	P	C	I.M	E.M	Exam
B23CIOE06	OE	3	0	0	3	30	70	3 Hrs.
COMPUTER NETWORKS								
(Offered by CIC)								
(Offered to CE, ECE, EEE & ME)								
Pre-requisites: Data Structures, Operating Systems.								
Course Objectives: Students are expected to								
1.	Understand the different types of networks							
2.	Discuss the software and hardware components of a network							
3.	Develop an understanding the principles of computer networks.							
4.	Familiarize with OSI model and the functions of layered structure.							
5.	Explain networking protocols, algorithms and design perspectives.							
Course Out Comes: At the end of the course students will be able to								
S. No	OUT COME							Knowledge Level
1.	Explain different types of networks, protocols, and reference models used in data communication.							K2
2.	Demonstrate physical and data link layer mechanisms such as framing, error control, and flow control in data communication.							K3
3.	Apply networking concepts and protocols to analyze and solve communication problems.							K3
4.	Use TCP and UDP protocols to manage data transmission between applications.							K3
5.	Apply application layer functionalities to facilitate communication between end-user applications over a network.							K3
SYLLABUS								
UNIT-I (10 Hrs)	Introduction: Types of Computer Networks: Broadband Access Networks, Mobile and Wireless Access Networks, Content Provider Networks, Transit networks, Enterprise Networks, Network technology from local to global, Network Protocols: Design Goals, Protocol Layering, Connections and Reliability, Service Primitives, The Relationship of Services to Protocols. Reference Models: The OSI Reference Model, The TCP/IP Reference Model, A Critique of the OSI Model and Protocols, A Critique of the TCP/IP Reference Model and Protocols.							
UNIT-II (10 Hrs)	The Physical Layer: Guided Transmission Media: Persistent Storage, Twisted Pairs, Coaxial Cable, Power Lines, Fiber Optics.							
	The Data Link Layer: Data Link Layer Design Issues: Services Provided to the Network Layer, Framing Error Control, Flow Control, Error Detection and Correction, Elementary Data							

	Link Protocols, Improving Efficiency. The Medium Access Control Sublayer: The Channel Allocation Problem, Multiple Access Protocols, Ethernet.
UNIT-III (10 Hrs)	The Network Layer: Network Layer Design Issues, Routing Algorithms In a Single Network, Traffic Management at The Network Layer: The Need for Traffic Management: Congestion, Approaches To Traffic Management, Internetworking, The Network Layer In The Internet, The IP Version 4 Protocol, IP Addresses, IP Version 6, Internet Control Protocols, Label Switching and MPLS, OSPF, BGP, Internet Multicasting.
UNIT-IV (10 Hrs)	The Transport Layer: The Transport Service: Services Provided To The Upper Layers, Transport Service Primitives, Berkeley Sockets, An Example Of Socket Programming, Elements Of Transport Protocols: Addressing, Connection Establishment, Connection Release, Error Control And Flow Control, Multiplexing, Crash Recovery, Congestion Control, The Internet Transport Protocols: UDP, The Internet Transport Protocols: TCP, Introduction To TCP, The TCP Service Model, The TCP Protocol, The TCP Segment Header, TCP Connection Establishment, TCP Connection Release.
UNIT-V (10 Hrs)	The Application Layer: Electronic Mail: Architecture and Services, The User Agent, Message Formats, Message Transfer, Final Delivery, The World Wide Web: Architectural Overview, Static Web Objects, Dynamic Web Pages and Web Applications, HTTP and HTTPS, Web Privacy, Content Delivery: Content and Internet Traffic, Server Farms and Web Proxies, Content Delivery Networks, Peer-To-Peer Networks, Evolution of The Internet.
TEXTBOOK: Estd. 1980 AUTONOMOUS	
1.	Andrew Tanenbaum, Feamster Wetherall, Computer Networks, 6th Edition, Global Edition, Pearson Higher Ed, 2021
REFERENCE BOOKS:	
1.	Behrouz A. Forouzan, Data Communications and Networking, 5th Edition, McGraw Hill Publication, 2017.
2.	James F. Kurose, Keith W. Ross, "Computer Networking: A Top-Down Approach", 6th edition, Pearson, 2019.
3.	YouluZheng, ShakilAkthar, "Networks for Computer Scientists and Engineers", Oxford Publishers, 2016.
E-Resources:	
1.	https://nptel.ac.in/courses/106105183/25
2.	http://www.nptelvideos.in/2012/11/computer-networks.html
3.	https://nptel.ac.in/courses/106105183/

Course Code	Category	L	T	P	C	C.I.E.	S.E.E.	Exam
B23ECOE03	OE	3	--	--	3	30	70	3 Hrs.
OPTICAL COMMUNICATION								
(Offered by ECE)								
(Offered to AIDS, AIML, CE, CIC, CSBS, CSE, CSG, CSIT, EEE, IT & ME)								
Course Objectives: Students are expected								
1.	To learn the basic elements of optical fiber transmission link, fiber modes, configurations and structures.							
2.	To understand the different kind of losses, signal distortion, Modes of fibers							
3.	To learn the various optical sources, materials and fiber splicing.							
4.	To learn the fiber optical receivers and noise performance in photo detector.							
5.	To learn link budget, WDM.							
Course Outcomes: After completion of the course, the student will be able to								
S.No	Outcome							Knowledge Level
1.	Understand the optical fiber communication link, structure, propagation and transmission properties of an optical fiber.							K3
2.	Design optical fiber experiments, and learn how to calculate electromagnetic modes in waveguides, the amount of light lost going through an optical system, dispersion of optical fibers.							K3
3.	Choose the optical cables for better communication with minimum losses							K3
4.	Use different types of photo detectors and optical test equipment to analyze optical fiber and light wave systems.							K3
5.	Design, build, and demonstrate optical fiber Link power budget calculations.							K3
SYLLABUS								
UNIT-I (10Hrs)	Overview of optical fiber communication: Advantages of optical fiber communications, Ray theory transmission, Total Internal Reflection, Acceptance angle, Numerical Aperture, V-number, Step Index fibers, Graded Index fibers, Related problems.							
UNIT- II (10 Hrs)	Fiber materials: Glass, Halide, Active glass, Plastic optical fibers. Attenuation, Absorption, Scattering and Bending losses, Material dispersion, Intermodal dispersion, Pulse broadening in Graded index fiber, Related problems.							
UNIT- III (10 Hrs)	Optical fiber Connectors- Connector types, Single mode fiber connectors, Fiber Splicing techniques, Splicing single mode fibers, Fiber alignment and single mode fiber joints.							

UNIT-IV (10 Hrs)	Optical sources -LEDs, Quantum efficiency, Modulation, Injection Laser Diodes- Modes, Reliability of LED & ILD, Optical detectors- Physical principles of PIN and APD, Comparison of Photo detectors.
UNIT- V (10 Hrs)	Source to fiber power launching - Power coupling, Laser diode to fiber coupling, Optical receiver operation, Receiver configuration, Link power budget, Rise time budget, WDM, Eye pattern.
Textbooks:	
1.	Fiber Optic Communications Technology – D.K. Mynbaev and Lowell L. Scheiner, Pearson Education, 2009.
2.	Optical Fiber Communications – Gerd Keiser, Mc Graw-Hill International edition, 4th Edition, 2000.
Reference Books:	
1.	Optical Fiber Communications – John M. Senior, PHI, 3rd Edition.
2.	Fiber Optic Communications – Joseph C. Palais, 4th Edition, Pearson Education, 2004.
e-Resources	
1.	https://www.thefoa.org/tech/ref/basic/fiber.html
2.	https://media.wiley.com/product_ancillary/17/04705051/DOWNLOAD/Fiber_Optic_Communication_Systems.pdf



Course Code	Category	L	T	P	C	C.I.E.	S.E.E.	Exam
B23ECOE04	OE	3	--	--	3	30	70	3 Hrs.
FUNDAMENTALS OF VLSI DESIGN								
(Offered by ECE)								
(Offered to AIDS, AIML, CE, CIC, CSBS, CSE, CSG, CSIT, EEE, IT & ME)								
Course Objectives: Students are expected								
1.	To introduce various fabrication steps of MOS transistors and their electrical properties							
2.	To implement the stick diagrams and layouts using CMOS/Bi-CMOS design rules.							
3.	To explain MOS technology interconnection as circuits, scaling models, static and dynamic designs.							
4.	To introduce Basic FPGA Architecture and testing methods of digital circuits							
Course Outcomes: After completion of the course, the student will be able to								
S. No	Outcome							Knowledge Level
1.	Analyze the Electrical properties and Fabrication processes of MOS circuits							K3
2.	Design the layouts of various MOS circuits by applying the concept of design rules.							K3
3.	Interpret the basic MOS circuit concepts							K2
4.	Interpret various scaling models and their impact on scaling of MOS circuits along with a few static and dynamic CMOS Designs							K2
SYLLABUS								
UNIT-I (10Hrs)	Introduction: Introduction to IC Technology, Fabrication process: NMOS, PMOS and CMOS. Ids versus Vds Relationships, Aspects of MOS transistor Threshold Voltage, MOS transistor Transconductance, Output Conductance and Figure of Merit. NMOS Inverter, Pull-up to Pull down Ratio for NMOS inverter driven by another NMOS Inverter, and through one or more pass transistors, Alternative forms of pull-up, The CMOS Inverter, Comparison between CMOS and Bi-CMOS technology.							
UNIT- II (10 Hrs)	MOS and Bi-CMOS Circuit Design Processes: MOS Layers, Stick Diagrams, Design Rules and Layout, General observations on the Design rules, 2µm Double Metal, Double Poly, CMOS/BiCMOS rules, Layout Diagrams of NAND and NOR gates and CMOS inverter.							
UNIT- III (10 Hrs)	Basic Circuit Concepts: Sheet Resistance, Sheet Resistance concept applied to MOS transistors and Inverters, Area Capacitance of Layers, Standard unit of capacitance, The Delay Unit, Inverter Delays, Propagation Delays, Wiring Capacitances, Choice of layers.							

UNIT-IV (10 Hrs)	Scaling of MOS Circuits: Scaling models, Scaling factors for device parameters, Limitations of Scaling on substrate doping, Miniaturization, Interconnect and contact Resistance, Subthreshold currents and current density.
UNIT- V (10 Hrs)	CMOS Combinational and Sequential logic circuit design: Static CMOS Design: Complementary CMOS, Ratioed logic, Pass Transistor logic- Design of logic gates. Dynamic CMOS Design: Basic principles, Static latches and registers, The bistability principle, SR- Flip flops, Multiplexer based latch.
Textbooks:	
1.	Essentials of VLSI Circuits and Systems By Kamran Eshraghian, Douglas and A. Pucknell and Sholeh Eshraghian, Prentice-Hall of India Private Limited, 2005 Edition
2.	Digital Integrated Circuits, Jan M. Rabaey, Anantha Chandrakasan and Borivoje Nikolic, 2nd edition, 2016
Reference Books:	
1.	FPGA Based System Design- Wayne Wolf, Pearson Education, 2004, Technology and Engineering
2.	CMOS Digital Integrated Circuits Analysis and Design, Sung-Mo Kang, Yusuf Leblebici, Tata McGraw Hill Education, 2003.
e-Resources	
1.	https://www.engineersgarage.com/vlsi-technology-an-overview/
2.	https://www.tutorialspoint.com/vlsi_design/vlsi_design_digital_system.htm

Course Code	Category	L	T	P	C	C.I.E.	S.E.E.	Exam
B23ECOE05	OE	3	--	--	3	30	70	3 Hrs.
ELECTRONIC MEASUREMENTS AND INSTRUMENTATION								
(Offered by ECE)								
(Offered to AIDS, AIML, CE, CIC, CSBS, CSE, CSG, CSIT, IT & ME)								
Course Objectives: Students are expected to								
1.	Select the instrument to be used based on the requirements.							
2.	Understand the design of oscilloscopes for different applications.							
3.	Understand the principle of operation and working of various types of bridges for measurement of parameters							
4.	Understand and analyze different signal generators and analyzers.							
Course Outcomes: After completion of the course, the student will be able to								
S.No	Outcome							Knowledge Level
1.	Evaluate basics of measurement systems, principle of basic meter.							K4
2.	Examining a signal / waveform with different oscillators.							K3
3.	Use bridges of many types and measure appropriate parameters.							K3
4.	Evaluate how a signal can be generated using different types of meters.							K4
5.	Design different transducers for measurement of different parameters.							K3
SYLLABUS								
UNIT-I (10Hrs)	Measuring Instruments: Introduction, Measurement standards, Performance characteristics of instruments, Static characteristics, Accuracy, Resolution, Precision, expected value, Error, Sensitivity. Errors in Measurement, Types of static errors- Gross errors, systematic errors, Instrumental errors, Observational errors, Random errors, Sources of error, Statistical analysis, Dynamic Characteristics-speed of response, Fidelity, Lag and Dynamic error. DC Voltmeters, AC voltmeters, True RMS responding voltmeter.							
UNIT- II (10 Hrs)	Oscilloscopes: CRT features, Block diagram of CRO, vertical amplifier, horizontal deflection system, sweep, trigger pulse, delay line. Dual beam CRO, Dual trace oscilloscope, sampling oscilloscope, digital storage oscilloscope.							
UNIT- III (10 Hrs)	Bridges: Measurement of inductance- Maxwell's bridge, Anderson bridge. Measurement of capacitance - Schearing Bridge, Wheatstone bridge, Wien Bridge, Errors and precautions in using bridges.							
UNIT-IV	Signal Generator: Introduction, fixed frequency AF oscillator, variable frequency AF oscillator, Basic Standard signal generator, AF sine and square wave signal generator,							

(10 Hrs)	Function Generators, Square and Pulse Generator, Sweep frequency generator.
UNIT- V (10 Hrs)	Transducers: Introduction, Types of Transducers, Resistive Transducer, Strain gauges, Piezo electric transducer, Linear Variable Differential Transducer.
Textbooks:	
1.	Electronic instrumentation, second edition - H.S.Kalsi, Tata McGraw Hill, 2004.
2.	Modern Electronic Instrumentation and Measurement Techniques–A.D. Helfrick and, D.W. Cooper, PHI, 5th Edition, 2002.
Reference Books:	
1.	Electronic Instrumentation & Measurements -David A.Bell, PHI, 2nd Edition, 2003.
2.	Electronic Test Instruments, Analog and Digital Measurements-Robert A.Witte, Pearson Education, 2nd Ed., 2004.
e-Resources	
1.	https://nptel.ac.in/courses/108/105/108105153/)
2.	https://onlinecourses.nptel.ac.in/noc19_ee44/preview
3.	https://www.youtube.com/watch?v=LM66kmlP_74



SRKR
ENGINEERING COLLEGE
AUTONOMOUS

Course Code	Category	L	T	P	C	C.I.E.	S.E.E.	Exam
B23EEOE05	OE	3	--	--	3	30	70	3 Hrs.
AUTONOMOUS VEHICLE SYSTEMS								
(Offered by EEE)								
(Offered to AIDS, AIML, CE, CIC, CSBS, CSE, CSG, CSIT, ECE, IT &ME)								
Course Objectives: Students will								
1.	Introduce the core principles of autonomous vehicles (AV), including their evolution, architecture, and fundamental kinematics/dynamics.							
2.	Explore sensor technologies and algorithms that enable AVs to interpret their surroundings.							
3.	Model vehicle behavior and implement control strategies for safe navigation.							
4.	Develop decision-making frameworks for complex driving scenarios.							
5.	Apply AV concepts in practice using the Robot Operating System (ROS).							
Course Outcomes: At the end of the course, the students will be able to								
S.No	Outcome							Knowledge Level
1.	Understand AV basics, autonomy levels, and apply AV kinematics.							K3
2.	Learn sensor technologies and how AVs perceive environments.							K4
3.	Model vehicle dynamics and control for stable driving.							K4
4.	Design decision-making systems for smart autonomous behavior.							K5
5.	Implement AV algorithms using ROS 2 and simulations.							K3
SYLLABUS								
UNIT-I (10Hrs)	Vehicle Autonomy Essentials: Introduction to AVs, historical evolution, SAE autonomy levels, AV architecture—Sensors, perception, planning, control and decision-making, Reference frames, coordinate systems, vehicle kinematics and dynamics, Transformation techniques – rotation matrices, homogeneous transforms, 3D representations.							
UNIT-II (10 Hrs)	Vehicle Dynamics and Motion Control: Longitudinal and Lateral dynamics, Dynamic bicycle model, Actuation Systems—Steering system dynamics, Brake-by-wire and throttle control, State-space representations, Motion Control, Control Objectives—Path tracking accuracy, Stability criteria (yaw rate, slip angle), Ride comfort metrics, Control Methods— PID control for basic maneuvers.							
UNIT-III (10 Hrs)	Perception and Sensor Systems: Sensors: LiDAR, Radar, Cameras, IMU, Ultrasonics, GNSS, Sensor fusion, State estimation—Kalman Filtering, Simultaneous Localization and Mapping (SLAM), object tracking, Visual odometry, LiDAR-based Mapping.							

UNIT-IV (10 Hrs)	Planning and Intelligent Decision-Making: Planning Hierarchy—Behavior planning vs motion planning, Scenario-based requirements, Rule-Based Systems—Finite State Machines (FSM) for driving behaviors, Decision trees for maneuver selection, Learning-Based Systems—Markov Decision Processes (MDP) framework, Deep Q-Networks (DQN) for driving policies, Reward Engineering—Safety metrics, Comfort parameters, Efficiency objectives.
UNIT-V (10 Hrs)	Robotic Operating Systems (ROS): ROS ecosystem, ROS - 2 Architecture, Package Management & Workspace Organization, ROS - 2 Communication—Publisher, Subscriber, Topic, Services, Actions, URDF and Robot Modeling, Simulation with Gazebo, RViz, Foxglove, Sensors integration with ROS2, ROS_control for drive-by-wire systems, ROS2 driver interfaces for motor controllers, Real-time topics: /odom, /tf, /sensor_data—/lidar/points, /cmd_vel, /gps/fix, /camera/info.
Textbooks:	
1.	Siegwart, R., Nourbakhsh, I.R., & Scaramuzza, D. Introduction to Autonomous Mobile Robots, MIT Press; 2nd Edition, 2011, ISBN: 978-0262015356.
2.	Paden, B., Čáp, M., Yong, S.Z., Yershov, D., & Frazzoli, E. A Survey of Motion Planning and Control Techniques for Self-Driving Urban Vehicles, Foundations and Trends® in Robotics, Now Publishers; 2016, ISBN: 978-1680831146.
Reference Books:	
1.	Maurer, M., Gerdes, J.C., Lenz, B., & Winner, H. Autonomous Driving: Technical, Legal and Social Aspects, Springer; 1st Edition, 2016, ISBN: 978-366248845.
2.	Sutton, R.S., & Barto, A.G. Reinforcement Learning: An Introduction, MIT Press; 2nd Edition, 2018, ISBN: 978-0262039246.
3.	Rajamani, R. Vehicle Dynamics and Control, Springer; 2nd Edition, 2012, ISBN: 978-1461414322.
4.	Ramkumar, L., & Kumar, A. ROS2 in 5 Days: Programming Robots with ROS2, ConstructSim; 2022.
5.	Fairchild, B. & Harman, J. Programming Robots with ROS: A Practical Introduction to the Robot Operating System, O'Reilly Media; 1st Edition, 2016., ISBN: 978-1449323899
e-Resources:	
1.	https://nptel.ac.in/courses/112106298
2.	https://www.udemy.com/course/applied-control-systems-for-engineers-2-uav-drone-control/?couponCode=KEEPLEARNING

Course Code	Category	L	T	P	C	C.I.E.	S.E.E.	Exam
B23EEOE06	OE	3	--	--	3	30	70	3 Hrs.
ELECTRICAL CIRCUIT ANALYSIS								
(Offered by EEE)								
(Offered to AIDS, AIML, CE, CIC, CSBS, CSE, CSG, CSIT, ECE, IT &ME)								
Course Objectives: Students will learn about								
1.	The various techniques of circuit analysis using resistive circuits.							
2.	The electrical circuits theorems.							
3.	The fundamentals of AC circuit analysis.							
4.	The magnetically coupled and 3 – phase circuits.							
5.	The two port networks and Resonance phenomenon in electrical circuits.							
Course Outcomes: At the end of the course, the students will be able to								
S.No.	Outcome							Knowledge Level
1.	Analyze DC resistive circuits with network analysis techniques.							K4
2.	Apply Network theorems for the analysis of electrical networks.							K3
3.	Analyze AC circuits using Phasor, Impedance and Complex Power concepts.							K4
4.	Analyze Magnetically coupled circuits and balanced three phase circuits.							K4
5.	Analyze the Resonant circuits and Two port networks.							K4
SYLLABUS								
UNIT-I (10Hrs)	DC Circuit Analysis: Circuit elements – Passive elements: R, L & C and their v-i relations - Active Elements: Voltage and Current Sources (Dependent and Independent) Analysis of DC resistive circuits: Single node-pair and single loop circuits, Source combinations, Resistances combinations, Voltage and Current division rules, star/delta transformation, Source transformation, Nodal and Mesh analysis techniques.							
UNIT-II (10 Hrs)	Network Theorems (DC Excitation): Superposition theorem - Thevenin's and Norton's theorems- Maximum power transfer theorem- Reciprocity theorem.							
UNIT-III (10 Hrs)	Single Phase AC Circuit Analysis: Characteristics of Sinusoids – Sinusoidal Steady State analysis – Phasor concept – Phasor voltage – current relations for R, L & C elements - Impedance and Admittance concepts - Nodal and Mesh Analysis - Phasor diagrams - Instantaneous power and Average Power - RMS values for periodic voltage and current waveforms - Complex Power.							

UNIT-IV (10 Hrs)	Coupled Circuits and Three Phase Circuit Analysis: Coupled Circuits: Concept of self and mutual inductance, Dot convention - Coefficient of coupling – Analysis of coupled circuits in Sinusoidal Steady state. Three Phase Circuit Analysis: Advantages of three phase circuits-Balanced Three Phase circuits – Line and Phase quantities in star/delta connections - Analysis of three phase balanced circuits, Power in 3-phase balanced circuits.
UNIT-V (10 Hrs)	Resonance and Two Port Networks: Resonance: Resonance phenomenon - RLC Series Resonance - Characteristics of a series resonant circuit– Bandwidth - Expressions for half power frequencies - Quality Factor - Selectivity. Two port networks: Z, Y, H, ABCD – Parameters of two port Networks, two port parameters for resistive Networks, Relationships between Parameter sets.
Textbooks:	
1.	Engineering Circuits Analysis, Jack Kemmerly, William Hayt and Steven Durbin, Tata Mc Graw Hill Education, 2020, 9th edition.
2.	Network Analysis, M.E. Van Valkenberg, Pearson Education, 2019, Revised Third Edition.
Reference Books:	
1.	Fundamentals of Electrical Circuits, Charles K Alexander and Mathew N.O Sadiku, Mc Graw Hill Education (India), 2022, 7th Edition.
2.	Schaum's Outline of Electric Circuits, by Mahmood Nahvi, Joseph Edminister, McGraw Hill; 7th edition (31 December 2017).
e-Resources:	
1.	https://nptel.ac.in/courses/108105159
2.	https://nptel.ac.in/courses/117106108

Course Code	Category	L	T	P	C	C.I.E.	S.E.E.	Exam
B23ITOE03	OE	3	--	--	3	30	70	3 Hrs.
PRINCIPLES OF DATABASE MANAGEMENT SYSTEMS								
(For IT)								
(Offered to CE, ECE, EEE & ME)								
Course Objectives: This course is designed to:								
1.	Introduce database management systems and to give a good formal foundation on the relational model of data.							
2.	Introduce the concepts of SQL							
3.	Demonstrate the principles behind systematic database design approaches by covering conceptual design, logical design through normalization							
4.	Provide overview of the transaction management							
Course Outcomes: Upon the completion of the course students will be able to:								
S.No	Outcome							Knowledge Level
1.	Understand the basic concepts of the database management system							K2
2.	Design databases by using E-R models							K3
3.	Apply SQL Concepts to create, manipulate, retrieve databases							K3
4.	Apply Normalization Concepts to refine relational databases							K3
5.	Understand the concepts of Transaction Management Systems							K2
SYLLABUS								
UNIT-I	Introduction: Database system, Characteristics (Database Vs File System), Database Users, Advantages of Database systems, Database applications.Brief introduction of different Data Models; Concepts of Schema, Instance and data independence; Three tier schema architecture for data independence; Database system structure, environment, Centralized and Client Server architecture for the database							
UNIT-II	Entity Relationship Model: Introduction, Representation of entities, attributes, entity set, relationship, relationship set, constraints, sub classes, super class, inheritance, specialization, generalization using ER Diagrams. Relational Model: Introduction to relational model, concepts of domain, attribute, tuple, relation, importance of null values, constraints (Domain, Key constraints, integrity constraints) and their importance.							

UNIT-III	BASIC SQL: Simple Database schema, data types, table definitions (create, alter), different DML operations (insert, delete, update). SQL: Basic SQL querying (select and project) using where clause, arithmetic & logical operations, SQL functions(Date and Time, Numeric, String conversion).Creating tables with relationship, implementation of key and integrity constraints, nested queries, sub queries, grouping, aggregation, ordering, implementation of different types of joins, view(updatable and nonupdatable)
UNIT-IV	Schema Refinement (Normalization): Purpose of Normalization or schema refinement, concept of functional dependency, normal forms based on functional dependency Lossless join and dependency preserving decomposition, (1NF, 2NF and 3 NF), Boyce-Codd normal form (BCNF)
UNIT-V	Transaction Concept: Transaction State, ACID properties, Concurrent Executions, Serializability, Recoverability, lock based (2-Phase,Strict 2-phase) concurrency protocols. Deadlocks(Detection and Prevention).
Textbooks:	
1.	Database Management Systems, 3rd edition, Raghurama Krishnan, Johannes Gehrke, TMH
2.	Database System Concepts,5th edition, Silberschatz, Korth, Sudarsan, TMH
Reference Books:	
1.	Introduction to Database Systems, 8thedition, C J Date, Pearson
2.	Database Management System, 6th edition, Ramez Elmasri, Shamkant B. Navathe, Pearson
3.	Database Principles Fundamentals of Design Implementation and Management, 10 th edition, Corlos Coronel, Steven Morris, Peter Robb, Cengage Learning, 2022

Course Code	Category	L	T	P	C	C.I.E.	S.E.E.	Exam
B23MEOE07	OE	3	--	--	3	30	70	3 Hrs.
TOTAL QUALITY MANAGEMENT								
(Offered by ME)								
(Offered to AIDS, AIML, CE, CSIT, CSBS, CSD, CSE, CIC, ECE, EEE & IT)								
Course Objectives: The student will acquire the knowledge to								
1.	Introduce the fundamentals of TQM and the role of process quality and statistical control in business performance.							
2.	Emphasize customer satisfaction, loyalty, and benchmarking practices for continuous improvement.							
3.	Explain how to organize and transition a company toward TQM using systems thinking and productivity tools.							
4.	Teach the concept and classification of quality costs and their role in quality management.							
5.	Provide knowledge of ISO 9000 standards and the certification process for global quality assurance.							
Course Outcomes: At the end of the course student will be able to								
S.No	Outcome							Knowledge Level
1.	Apply process control tools like control charts and acceptance sampling, and understand TQM principles.							K3
2.	Evaluate customer-focused strategies and implement benchmarking processes to enhance quality.							K3
3.	Implement organizational changes for TQM and assess productivity improvements through quality circles and reengineering.							K3
4.	Apply quality cost data to drive managerial decisions and quality improvements.							K3
5.	Apply ISO 9000 certification procedures and assess their impact on business quality systems.							K3
SYLLABUS								
UNIT-I (10Hrs)	INTRODUCTION: The concept of TQM, Quality and Business performance, attitude and involvement of top management, communication, culture and management systems. Management of Process Quality: Definition of quality, Quality Control, a brief history, Product Inspection vs, Process Control, Statistical Quality Control, Control Charts and Acceptance Sampling.							
UNIT-II (12 Hrs)	CUSTOMER FOCUS AND SATISFACTION: The importance of customer satisfaction and loyalty- Crating satisfied customers, Understanding the customer needs, Process Vs. Customer, internal customer conflict, quality focus, Customer Satisfaction, role of							

	Marketing and Sales, Buyer – Supplier relationships. Bench Marketing: Evolution of Bench Marketing, meaning of Bench marketing, benefits of bench marketing, the bench marketing process, pitfalls of bench marketing.
UNIT-III (10 Hrs)	ORGANIZING FOR TQM: The systems approach, organizing for quality implementation, making the transition from a traditional to a TQM organizing, Quality Circles. Productivity, Quality and Reengineering: The leverage of Productivity and Quality, Management systems Vs. Technology, Measuring Productivity, Improving Productivity Re-engineering.
UNIT-IV (8 Hrs)	THE COST OF QUALITY: Definition of the Cost of Quality, Quality Costs, Measuring Quality Costs, use of Quality Cost Information, Accounting Systems and Quality Management.
UNIT-V (8 Hrs)	ISO 9000: Universal Standards of Quality: ISO around the world, The ISO 9000 ANSI/ASQCQ-Series Standards, benefits of ISO 9000 certification, the third-party audit, Documentation ISO 9000 and services, the cost of certification implementing the system
Textbooks:	
1.	Total Quality Management / Joel E.Ross/Taylor and Franscis Limited
2.	Total Quality Management/P.N.Mukherjee/PHI
Reference Books:	
1.	Beyond TQM / Robert L.Flood
2.	Statistical Quality Control / E.L. Grant / McGraw Hill.
3.	Total Quality Management- A Practical Approach/H. Lal
4.	Quality Management/KanishkaBedi/Oxford University Press/2011
5.	Total Engineering Quality Management/Sunil Sharma/Macmillan
e-Resources	
1.	https://nptel.ac.in/courses/110104080
2.	https://onlinecourses.nptel.ac.in/noc20_mg34/preview

Course Code	Category	L	T	P	C	C.I.E.	S.E.E.	Exam
B23MEOE08	OE	3	--	--	3	30	70	3 Hrs.
SMART MANUFACTURING								
(Offered by ME)								
(Offered to AIDS, AIML, CE, CSIT, CSBS, CSD, CSE, CIC, ECE, EEE & IT)								
Course Objectives:								
1.	Apply the concepts of smart and circular manufacturing, analyzing challenges such as scalability and resource optimization, and evaluating how smart machines and sensors improve operational efficiency in industrial environments.							
2.	Apply the design principles of Cyber-Physical Systems (CPS) and evaluate the 5C architecture to optimize smart factory operations, integrating digital twin technology and AI/ML for enhanced decision-making and predictive maintenance.							
3.	Explore IoT communication protocols and evaluate the connectivity infrastructure required to enable seamless integration, automation, and data exchange in industrial environments.							
Course Outcomes								
S.No	Outcome							Knowledge Level
1.	Apply the principles of smart manufacturing, Industry 4.0, and Industry 5.0 by analyzing their evolution, stages, and the benefits they bring to modern manufacturing systems.							K3
2.	Apply the knowledge of smart sensors, subsystems, and interfaces used in IoT-based manufacturing, and describe their integration to enhance production systems.							K3
3.	Apply the 5C architecture of Cyber-Physical Systems (CPS) to explain the functioning of smart manufacturing systems, including PHM-based design, and distinguish between traditional factories and Industry 4.0 factories based on CPS implementation.							K3
4.	Apply digital twin implementation concepts in manufacturing and demonstrate the use of AI/ML for predictive maintenance to improve operational efficiency.							K3
5.	Apply an understanding of IoT network layers and protocols used in Industry 4.0 applications, and differentiate between their roles in enabling industrial automation.							K3
SYLLABUS								
UNIT-I (10Hrs)	CONCEPTS OF SMART MANUFACTURING: Definition and key characteristics of smart manufacturing, Corporate adaptation processes, manufacturing challenges, challenges vs technologies, Overview of circular manufacturing, Stages in smart manufacturing. Minimizing Six big losses in manufacturing with Industry 4.0, Industry 5.0, and their benefits							

UNIT-II (10 Hrs)	SMART MACHINES AND SMART SENSORS: Concept and Functions of a Smart, Machine Salient features and Critical Subsystems of a Smart Machine, Smart sensors; smart sensors ecosystem, need, benefits and applications of sensors in industry, Sensing for Manufacturing Process in IIoT, Block Diagram of a IoT Sensing Device, Sensors in IIoT Applications, Smart Machine Interfaces
UNIT-III (10 Hrs)	ARCHITECTURE OF CYBER-PHYSICAL SYSTEM (CPS): Functions of CPS, 5C Architecture; Smart Connection Level, Data-to- Information Level, Cyber Level, Cognition Level, Configuration Level. Design of PHM based CPS systems. Comparison of today's factory and Industry 4.0 factory by the implementation of 5C CPS architecture
UNIT-IV (10 Hrs)	DIGITAL TWIN: Introduction, applications of digital twins, impact zones of digital twins in manufacturing (factories/plants and OEMs), advantages of digital twins, basic steps of digital twin technology, Multiverse and AR and VR. MACHINE LEARNING (ML) AND ARTIFICIAL INTELLIGENCE (AI) IN MANUFACTURING: Introduction, benefits and applications of ML in industries, common approaches of ML; supervised and unsupervised, semi supervised and reinforced ML, Application of AI and ML for predictive maintenance.
UNIT-V (10 Hrs)	IOT CONNECTIVITY FOR INDUSTRY 4.0: Industrial communication requirement and its infrastructure, an overview of different types of networks, mesh network in industrial IoT, IoT protocols and the internet, TCP/IP (transmission control protocol/internet protocol) model, IoT connectivity standards: common protocols, application layer protocols, internet/network layer protocols, physical layer IoT protocols.
Textbooks:	
1.	"Industry 4.0 The Industrial Internet of Things" by Alasdair Gilchrist, Apress
2.	"Industrial Internet of Things, Cyber Manufacturing System" by Sabina Jeschke, Christian Brecher, Houbing Song Danda B. Rawat, Springer
Reference Books:	
1.	"The Fourth Industrial Revolution" by Klaus Schwab
2.	"Internet of Things: A Hands-On Approach" by Arshdeep Bahga& Vijay Madisetti
3.	"Cyber-Physical Systems: From Theory to Practice" by Rajeev Alur, Daniel D. Lee, and Insup Lee
e-Resources	
1.	https://www.coursera.org/learn/industry-4-0
2.	https://ocw.mit.edu/courses/electrical-engineering-and-computer-science/6-831-cyber-physical-systems-spring-2012/

Course Code	Category	L	T	P	C	I.M	E.M.	Exam
B23BSOE03	OE	3	--	--	3	30	70	3 Hrs.
MATHEMATICAL MODELLING FOR DATA SCIENCE (MMDS)								
(offered by- EM&H)								
(offered to All Branches)								
Course Objectives: Students are expected to:								
1	Learn Joint random variables, Markov process and its properties.							
2	Understand various queuing systems and their applications.							
3	Have an idea on Generation of random numbers and its applications in various domains.							
4	Know how to design and conduct experiments using ANOVA.							
5	Learn how to test the hypothesis for Non-Parametric data.							
Course Outcomes: After completion of the course, the student will be able to								
S. No.	Outcome							Knowledge Level
1	Find the Joint and conditional probabilities and apply them to a Stochastic process							K3
2	Identify various queuing models and find its solutions.							K3
3	Distinguish different simulation models and solve them							K3
4	Apply ANOVA techniques to the given data.							K3
5	Make use of testing of hypothesis and its applications to Non-Parametric data.							K3
SYLLABUS								
UNIT-I (12Hrs)	Stochastic Process: Random variables, Joint probability distribution, conditional probability distribution, marginal probabilities, concept of random process (stochastic process), Classification of process, Statistical independence, Markov chain, Stochastic matrix-properties (without proof). Transition probabilities: One-step & n-step transition probabilities.							
UNIT-II (10Hrs)	Queuing Models: - Queuing system, Characteristics, Transient and steady states, Probability distribution in queuing systems, Poisson process, Kendall's notation, Classification of queuing models - (M/M/1): (∞ /FCFS) and (M/M/1): (N/FCFS). Introduction to Networks of queues.							
UNIT-III (10Hrs)	Simulation: - Introduction to simulation, phases of simulation, systems and system environment, Components of a system, discrete and continuous systems, Steps in simulation study, Generation of random numbers, Monte-Carlo simulation. Simulation applications- Applications to inventory control, Queuing model.							
UNIT-IV (10 Hrs)	ANOVA and its applications: Introduction, Analysis of Variance one-way and two-way							

	classification. Applications of ANOVA in various fields.
UNIT-V (12 Hrs)	Non-parametric Inference: Comparison with parametric inference, Use of order statistics. Sign test, Wilcoxon signed rank test, Mann-Whitney test, Run test, Kolmogorov-Smirnov Test, Spearman and Kendall's test.
Text Books:	
1.	Higher Engineering Mathematics by B V Ramana, MC-Graw Hill Edn.(Unit-I)
2.	Operations Research, S.D. Sharma, Kedharnath and Ramnath.-(Unit-II and Unit-III)
3.	Fundamentals of Applied Statistics, S.C. Gupta & V.K. Kapoor, Sultan Chand & Son's. (Unit-IV)
4.	Non-Parametric Statistics, A Step-by-Step Approach, Gregory W. Corder & Dale I. Foreman, Wiley Publications. (Unit-V)
Reference Books:	
1.	Probability Theory and stochastic process for engineers, K N Hari Bhat, Anitha Sheela. K, Jayant Gangula, Pearson publishers.
2.	Operations Research by J.K Sharma,MACMILAN publications
3.	Discrete-event system simulation, Jerry bank, J.s. Carson, B L Nelson & David M.Nical, Prentice-Hall of India, (3rd Edition)
4.	Fundamentals of Statistics, S.C. Gupta, Himalayan Publishing House.
5.	Applied Multivariate Statistical Analysis, Richard. A. Johnson Dean. W. Wkhern. (6th Edition)
6.	Fundamentals of Mathematical Statistics, S.C. Gupta & V.K. Kapoor, Sultan Chand & Sons.